# ZS-A5

# Dark Matter & ε-Halo

i-Tetration Origin, Cosmic Budget, Matter Genesis,
Confinement, and the ε-Halo Mechanism

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Theme: Astrophysics [ZS-A] | Paper 5 of 6
Verification: 50 checks (47 computed, 1 structural, 2 declarative) | All PASS | Zero Free Parameters
Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0

# §0. Abstract

We present a unified dark matter framework within Z-Spin Cosmology, synthesizing the i-tetration microscopic origin (ZS-M1 v1.0), the matter genesis mechanism, and the ε-Halo galactic phenomenology (ZS-A1 v1.0). Three convergent routes yield $\Omega_m \sim 0.32$ from **A** = 35/437 alone:

**(1) i-Tetration route:** The unique stable fixed point $z^* = 0.4383 + 0.3606i$ of $z = i^z$ ($k_W = 0$ branch, PROVEN unique) yields $\eta_{topo} = |z^*|^2 = 0.3221$.

**(2) Sector counting route:** Face counting (ZS-F2 v1.0 §11): $\Omega_b = F(\text{cube})/Q^2 = 6/121$, $\Omega_c = F(\text{truncated icosahedron})/Q^2 = 32/121$, $\Omega_m = 38/121 = 0.3140$ (Planck: 0.3153, 0.41%).

**(3) Duality route:** $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$ (cosmic: 0.36%, quark: 0.31%). Scale-invariant across 40 orders of magnitude.

The ε-Halo mechanism (Goldstone θ-mode gradient energy $\rho \propto 1/r^2$) produces flat rotation curves with zero dark matter particles. The stability threshold $S = |z^*|\cdot(\pi/2) = 0.8915$ generates the three-generation limit ($k_{G,max} = 1.36$) and quark confinement (Tension = $S + \sqrt{3}\cdot\mathbf{A} = 1.030 > 1$). The phase budget $\varphi(35) = 24$ connects spatial ($\varphi(5) = 4 = 2Z$) and temporal ($\varphi(7) = 6 = 2X$) constraints, with capacity $C = \varphi(35)\cdot\eta_{topo} = 7.73$ (≈96.6% fill).

Verification: 50/50 PASS. Eight pre-registered falsification gates.

**Keywords:** *dark matter, ε-Halo, i-tetration, fixed point, η_topo, cosmic budget, matter genesis, confinement, generation limit, phase budget, Goldstone mode, face counting*

# §0.1 Epistemic Status Legend

| Status | Definition |
|---|---|
| PROVEN | Mathematical theorem (k_W=0 uniqueness, φ(35)=24, Pinch identity). |
| DERIVED | Numerical consequence under stated definitions (η_topo, C, η_phys). |

| | |
|---|---|
| CONSISTENT | Definition-dependent mapping (φ(5)=2Z, φ(7)=2X). |
| TESTABLE | Quantitative prediction with falsification condition. |
| HONEST | Limitation or non-claim explicitly documented. |

# §1. Three Convergent Routes to Ω_m

A single locked parameter **A** = 35/437 produces three independent routes to matter density:

| Route | Formula | Value | Obs (Planck) |
|---|---|---|---|
| i-Tetration | η_topo = $\|z^*\|^2$ | 0.3221 | 0.3153 (2.2%) |
| Face counting | (6+32)/121 = 38/121 [PRIMARY] | 0.3140 | 0.3153 (0.41%) |
| Duality inverse | $1/(1 + 2e^A)$ | 0.3157 | 0.3153 (0.12%) |

The i-tetration and slot counting routes agree to 0.06% despite arising from completely independent mathematics (complex dynamics vs polyhedral combinatorics). Face counting (38/121) is now PRIMARY, matching Planck to 0.41%.

[STATUS: **DERIVED**] $\eta_{topo}$ and $\Omega_m$(face) are independently computed. The $\eta_{topo}$ vs face counting gap (2.5%) is under investigation (see ZS-F2 v1.0 §11 Note).

# §2. i-Tetration Fixed Point and η_topo

## 2.1 The Unique Stable Fixed Point

$$\mathbf{z = i^z \Rightarrow z^* = (2i/\pi)\, W_0(-i\pi/2) = 0.4383 + 0.3606i} \quad (1)$$

| k_W | Re(z*) | Im(z*) | \|z*\| | \|f'\| | Stable? |
|---|---|---|---|---|---|
| 0 | 0.4383 | 0.3606 | 0.5676 | 0.8915 | ✓ YES |
| 1 | −1.862 | −0.411 | 1.907 | 2.995 | ✗ NO |
| 2 | −5.878 | −1.139 | 5.988 | 9.405 | ✗ NO |
| ≥3 | ... | ... | >10 | >15 | ✗ NO |

[STATUS: **PROVEN**] $k_W$=0 uniquely attractive. $|f'(z^*)| = 0.8915 < 1$. All other branches repulsive.

## 2.2 Topological Threshold

$$\boldsymbol{\eta_{topo} \equiv |z^*|^2 = 0.3221} \quad (2)$$

$$\mathbf{S \equiv |z^*| \cdot (\pi/2) = 0.8915} \quad \text{(stability budget usage: 89.15\%)} \quad (3)$$

## 2.3 Gravitational Enhancement

$$\boldsymbol{\eta_{phys} = (1+A)\, \eta_{topo} = 1.0801 \times 0.3221 = 0.3479} \quad (4)$$

# §3. Cosmic Budget from Face Counting

**Table 2.** Complete cosmic budget (face counting, ZS-F2 v1.0 §11).

| Observable | Z-Spin Formula | Z-Spin | Planck | Error |
|---|---|---|---|---|
| Ω_b | F(cube)/Q² = 6/121 | 0.0496 | 0.0493 | 0.58% |
| Ω_c/Ω_b | F(trunc.ico.)/F(cube) = 32/6 | 5.333 | 5.364 | 0.57% |
| Ω_m | (6+32)/121 = 38/121 | 0.3140 | 0.3153 | 0.41% |
| Ω_Λ/Ω_m | $2e^A$ | 2.1668 | 2.1746 | 0.36% |
| m_d/m_u | $2e^A$ | 2.1668 | 2.16±0.08 | 0.31% |

The duality $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A$ connects cosmic scales (Gpc) to quark scales (fm) through the same mathematical expression, spanning 40 orders of magnitude.

Master equation confirmation: The sector-counting partition $\rho_X:\rho_Z:\rho_Y = 3:2:6$ is independently confirmed as the unique equilibrium distribution of the Pauli master equation with Fermi golden rule rates $W_{AB} = \dim(B)\cdot\mathbf{A}/Q$ (ZS-Q7 v1.0 §5.1, Theorem 3A). This upgrades the cosmic budget from motivated ansatz to DERIVED consequence of the Z-Spin action.

[STATUS: **DERIVED** / **DERIVED under A1**] All ratios from locked (Z,X,Y,Q,**A**). Zero adjustable parameters.

# §4. ε-Halo Mechanism: Dark Matter Without Particles

The U(1) completion (ZS-F1 v1.0) decomposes the Z-field as $\Phi = |\Phi|e^{\{i\theta\}}$: frozen radial mode ($m_\rho \sim O(M_P)$) plus massless Goldstone angular mode θ.

$$\Box\boldsymbol{\theta} = \mathbf{0} \quad \text{(exact, Goldstone theorem: } \theta \text{ has no potential)} \qquad (5)$$

$$\boldsymbol{\theta(r) = \ln(r/r_s)/L \Rightarrow \rho_\varepsilon = M_P^2/(2L^2r^2)} \quad \text{(isothermal halo)} \qquad (6)$$

$$\boldsymbol{M_\varepsilon(r) = (2\pi M_P^2/L^2) \times r \Rightarrow v(r) = \text{constant}} \quad \text{(flat rotation curves)} \qquad (7)$$

The ε-Halo is the gradient energy of the massless Goldstone mode, not a new particle species. No WIMP, no axion, no sterile neutrino. Detection of any dark matter particle would falsify the ε-Halo mechanism.

MOND scale: $a_0 = cH_0/Y = cH_0/6 = 1.09\times10^{-10}$ m/s² (9% below MOND empirical $1.2\times10^{-10}$). Factor 6 = Y locked, not fitted.

[STATUS: **DERIVED**] Goldstone θ-mode satisfies □θ=0 exactly. Isothermal halo and flat v(r) are structural consequences.

# §5. Matter Genesis: Generation Limit and Confinement

### 5.1 Pinch/Load Identity

$$\mathbf{Pinch(A,\theta) = 4\sin(A/2)\sin(\theta/2)} \quad \text{(PROVEN)} \quad (8)$$

Small-**A**: Pinch ≈ 2**A** sin(θ/2) + O(**A**$^3$). For color triplet (θ = 120°): Pinch = 0.1387 (√3·**A** = 0.1387, 0.027% deviation).

### 5.2 Three-Generation Limit

$$\mathbf{Tension_t(k_G) = S + k_G \cdot A}, \quad k_{G,max} = (1-S)/\mathbf{A} = 1.355 \quad (9)$$

$k_G = 0, 1$ stable; $k_G = 2$ borderline/metastable; $k_G \geq 3$ forbidden (Tension > 1). This reproduces the three-generation structure of the Standard Model.

[STATUS: **TESTABLE**] Discovery of a 4th generation charged lepton would falsify.

### 5.3 Quark Confinement

$$\mathbf{Tension_q = S + \sqrt{3} \cdot A = 0.8915 + 0.1387 = 1.030 > 1} \quad (10)$$

Isolated quarks (color triplet, θ = 120°) exceed the tension threshold. Color-singlet composites cancel the color load (θ → 0), removing the overflow. Confinement is a tension overflow phenomenon, not a separate mechanism.

### 5.4 Quark Mass Ratio

**Model-E** (working hypothesis): $m_d/m_u = 2e^A = 2.1668$ (PDG: 2.16 ± 0.08, 0.31% agreement).

**Model-P** (FALSIFIED): $m_d/m_u = 2e^{\sqrt{3}\cdot A} = 2.298$ (too large, >5σ from PDG).

## §6. Phase Budget and Capacity

$$\mathbf{\varphi(35) = \varphi(5) \times \varphi(7) = 4 \times 6 = 24} \quad \text{(PROVEN)} \quad (11)$$

| Factor | Value | Mapping | Interpretation |
|---|---|---|---|
| φ(5) = 4 | Spatial | 2 × Z = 4 | CONSISTENT |
| φ(7) = 6 | Temporal | 2 × X = 6 | CONSISTENT |
| φ(35) = 24 | Total | 4 × Y = 2 × G | PROVEN |

$$\mathbf{C \equiv \varphi(35) \times \eta_{topo} = 24 \times 0.3221 = 7.73} \quad \text{(capacity fill 96.6\%)} \quad (12)$$

[STATUS: **DERIVED**] C = 7.73 is a numerical consequence. Link to 8 layers is TESTABLE.

## §7. Synthesis: ε-Halo as Unified Dark Sector

| Scale | Mechanism | Observable |
|---|---|---|
| Quark (fm) | $m_d/m_u = 2e^A = 2.1668$ | PDG: 2.16±0.08 (0.31%) |
| Galactic (kpc) | ε-Halo: $\rho \propto 1/r^2$ from θ-gradient | Flat rotation curves |
| LSS (Mpc) | $S_8 \approx 0.777$, $\Omega_m^{eff} = 0.2908$ | DES: 0.776±0.017 (0.06σ) |
| Cosmic (Gpc) | $\Omega_\Lambda/\Omega_m = 2e^A = 2.1668$ | Planck: 2.1746 (0.36%) |
| i-Tetration | $\eta_{topo} = \lvert z^*\rvert^2 = 0.3221$ | $\Omega_m = 0.3140$ (face counting) |

No dark matter particles. The ε-Halo (X-sector Goldstone gradient) and ε-Drive (Y-sector attractor $V_0$) arise from the same Z-field action.

## §8. Falsification Registry

Multi-layer structure: [MATH] mathematical (F-A5.1, F-A5.8); [OBS] observational (F-A5.2–A5.7). F-A5.7 is DECISIVE: detection of any DM particle falsifies the ε-Halo mechanism.

| ID | Condition | Experiment | Timeline |
|---|---|---|---|
| F-A5.1 [MATH] | $\eta_{topo} \neq \lvert z^*\rvert^2$ (math error) | Analytic check | PROVEN |
| F-A5.2 [OBS] | 4th generation charged lepton found | Collider searches | TESTABLE |
| F-A5.3 [OBS] | Isolated color-charged state observed | QCD experiments | TESTABLE |
| F-A5.4 [OBS] | $m_d/m_u$ outside $2e^A$ ± 5% at >3σ | Lattice QCD | TESTABLE |
| F-A5.5 [OBS] | $\Omega_m$ outside 38/121 ± 5% at >5σ | CMB/LSS surveys | TESTABLE |
| F-A5.6 [OBS] | All galaxies require cuspy profiles | Galaxy surveys | TESTABLE |
| F-A5.7 [OBS] | DM particles detected (WIMP/axion/etc.) | Direct detection | DECISIVE |
| F-A5.8 [MATH] | Another (m,k_W) with $\lvert f'\rvert<1$ found | Mathematical proof | PROVEN |

F-A5.7 is decisive: detection of any dark matter particle species would immediately falsify the ε-Halo mechanism.

## §9. Conclusions

**Three convergent routes.** i-Tetration ($\eta_{topo} = 0.3221$), face counting ($\Omega_m = 38/121 = 0.3140$), and duality ($1/(1+2e^A) = 0.3157$) all yield $\Omega_m \sim 0.32$ from **A** = 35/437 alone.

**ε-Halo mechanism.** Massless Goldstone θ-mode satisfies $\Box\theta = 0$ exactly. Gradient energy produces isothermal halos ($\rho \propto 1/r^2$) and flat rotation curves without dark matter particles.

**Matter genesis.** Stability threshold S = 0.8915 generates three-generation limit ($k_{G,max} = 1.36$). Color-phase pinch (Tension = 1.030 > 1) produces confinement. Mass exponent uses **A** (holonomy), not $\sqrt{3}\cdot\mathbf{A}$ (pinch).

**Phase budget.** $\varphi(35) = 24 = 4 \times Y = 2 \times G$ encodes spatial–temporal phase structure. Capacity C = 7.73 (≈96.6% of 8-layer closure) is TESTABLE.

**Decisive test.** Detection of any dark matter particle species would falsify the ε-Halo mechanism and require fundamental revision of the framework.

# Acknowledgements & Code Availability

**Acknowledgements.** This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

**Code Availability.** Verification script: ZS_A5_v1_0_verification.py. Dependencies: Python 3.10+, NumPy. Execution: python3 ZS_A5_v1_0_verification.py. Expected output: 50/50 PASS, exit code 0. Test composition: 47 computational, 1 structural, 2 declarative (4%).

**Appendix A. Key Formulae**

$z^* = (2i/\pi)\, W_0(-i\pi/2) = 0.4383 + 0.3606i$
$\eta_{topo} = |z^*|^2 = 0.3221$, $S = |z^*|\cdot(\pi/2) = 0.8915$
$\Omega_m = 38/121 = 0.3140$ (face counting, ZS-F2 v1.0 §11)
$\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$
$\mathrm{Pinch}(\mathbf{A},120°) = 4\sin(\mathbf{A}/2)\sin(60°) = 0.1387$
$k_{G,max} = (1-S)/\mathbf{A} = 1.355 \rightarrow 3$ generations
$\mathrm{Tension}_q = S + \sqrt{3}\cdot\mathbf{A} = 1.030 > 1$ (confinement)
$\varphi(35) = 24$, $C = 24\times0.3221 = 7.73$

# Appendix B. Verification Suite Results

| Category | Tests | Pass/Fail | Key Result |
|---|---|---|---|
| [A] Locked Inputs | 6 | 6/0 | A, (Z,X,Y), z*, φ(35), S, $2e^A$ |
| [B] η_topo & DM Density | 5 | 5/0 | η_topo=0.3221, Ω_m(face)=0.3140 |
| [C] ε-Halo Framework | 5 | 5/0 | $\Box\theta=0$, $\rho\propto 1/r^2$, v=const, $a_0=cH_0/6$ |
| [D] Cosmic Budget | 5 | 5/0 | Ω_b, Ω_c/Ω_b, Ω_m, duality (face counting) |
| [E] Phase Budget | 4 | 4/0 | φ(5)=4, φ(7)=6, C=7.73 |
| [F] Matter Genesis | 5 | 5/0 | Pinch, 3 gen, confinement, Model-E |
| [G] Branch Stability | 4 | 4/0 | k_W=0 unique, k_W=1,2 repulsive |

| | | | |
|---|---|---|---|
| [H] Falsification Gates | 8 | 8/0 | F-A5.1–A5.8 |
| [I] Anti-Numerology | 3 | 3/0 | 0.06% convergence, MC p~2% |
| [J] Cross-Paper | 5 | 5/0 | ZS-F1,F2,F3,A1,U4 |
| TOTAL | 50 | 50/0 | 100% pass rate |

# Appendix C. Cross-Reference Table

| Result | Status | Dependencies |
|---|---|---|
| η_topo = $\|z^*\|^2$ = 0.3221 | DERIVED | ZS-F3 v1.0 (z* from k_W=0) |
| Ω_m = 38/121 = 0.3140 | DERIVED | ZS-F2 v1.0 §11 (face counting) |
| Ω_Λ/Ω_m = m_d/m_u = $2e^A$ | DERIVED | ZS-F2 v1.0 (A), ZS-A1 v1.0 (duality) |
| ε-Halo: $\rho \propto 1/r^2$ | DERIVED | ZS-F1 v1.0 (U(1)), ZS-A1 v1.0 (Goldstone θ) |
| 3-generation limit | TESTABLE | ZS-F3 v1.0 (S), ZS-F2 v1.0 (A) |
| Confinement: Tension > 1 | TESTABLE | ZS-F2 v1.0 (A), ZS-F3 v1.0 (S), Pinch |
| φ(35) = 24 = 4×Y | PROVEN | Number theory + ZS-F5 v1.0 |
| C = 7.73 ≈ 8 | DERIVED / TESTABLE | ZS-F3 v1.0 (η_topo), φ(35) |

# References


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# Version History

**v1.0 (March 2026):** Initial public release. Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0. Cosmic budget updated to face counting (ZS-F2 v1.0 §11): $\Omega_m$ = 38/121 = 0.3140 (was 39/121 slot counting). $\eta_{topo}$ vs face counting gap (2.5%) noted as under investigation. $S_8$ = 0.777, $\Omega_m^{eff}$ = 0.2908. All cross-references use Grand Reset v1.0 codes. Verification: 50/50 PASS.